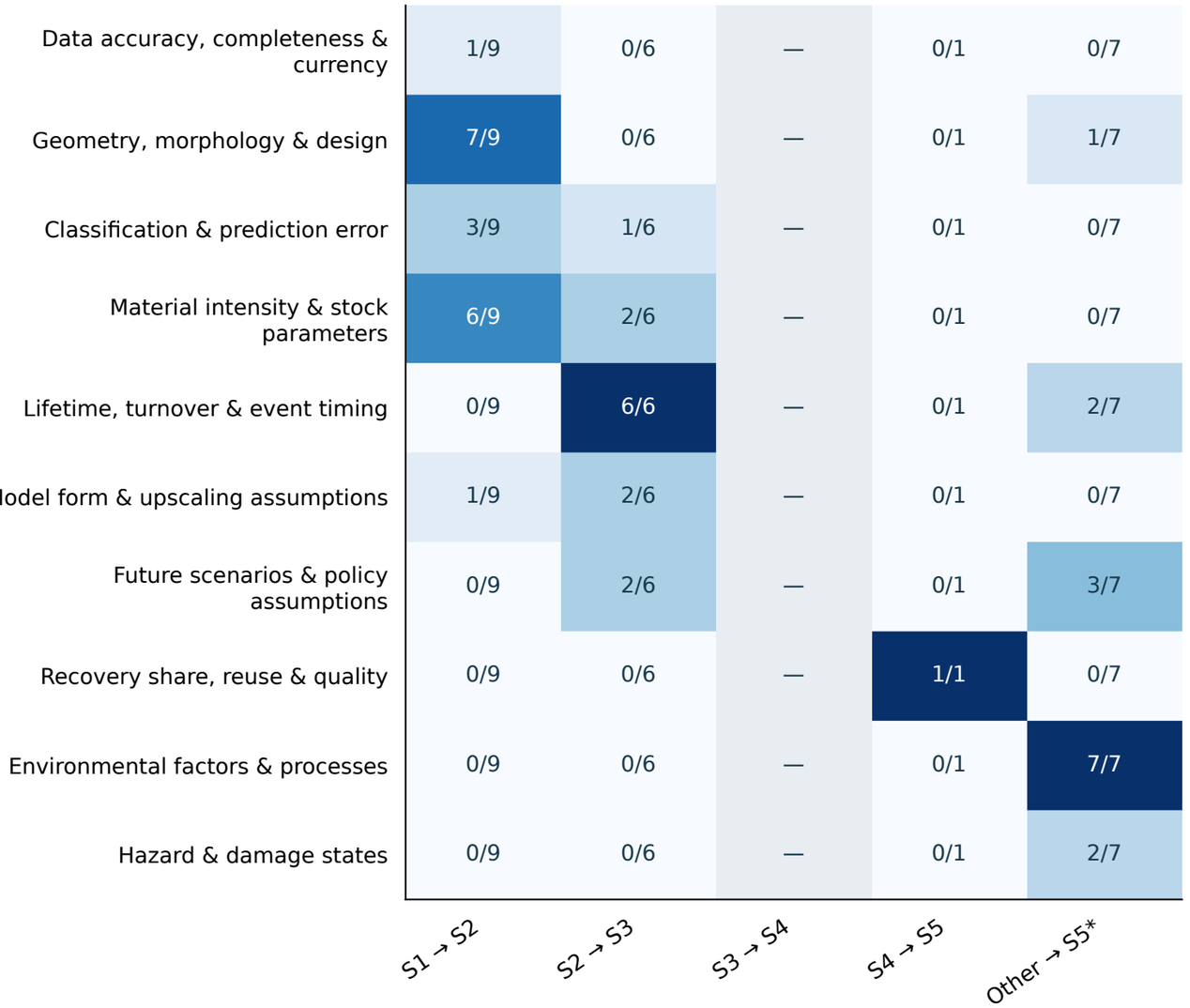
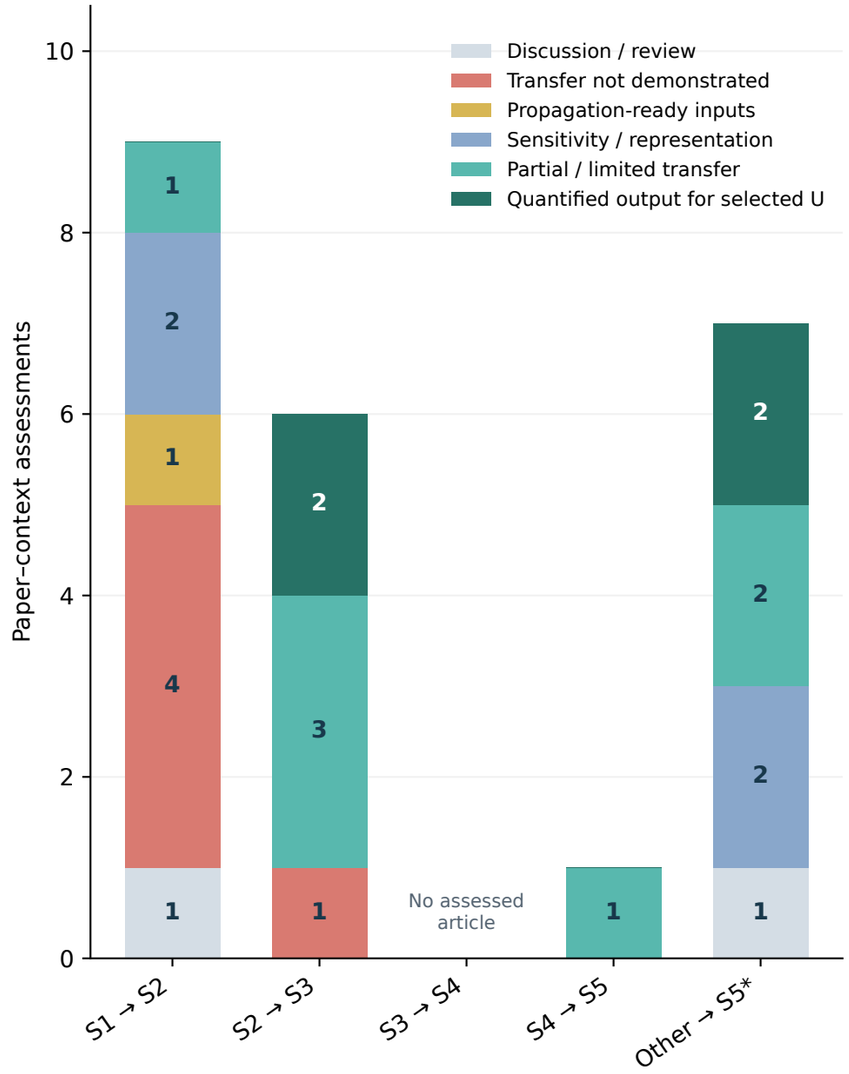
01 Stages × uncertainty

Recorded sources and continuity are different outcomes. Working set: 20 assessed papers; U10 excluded.

Source coverage
Papers / assessed context papers



Continuity evidence
One assessment per paper–context



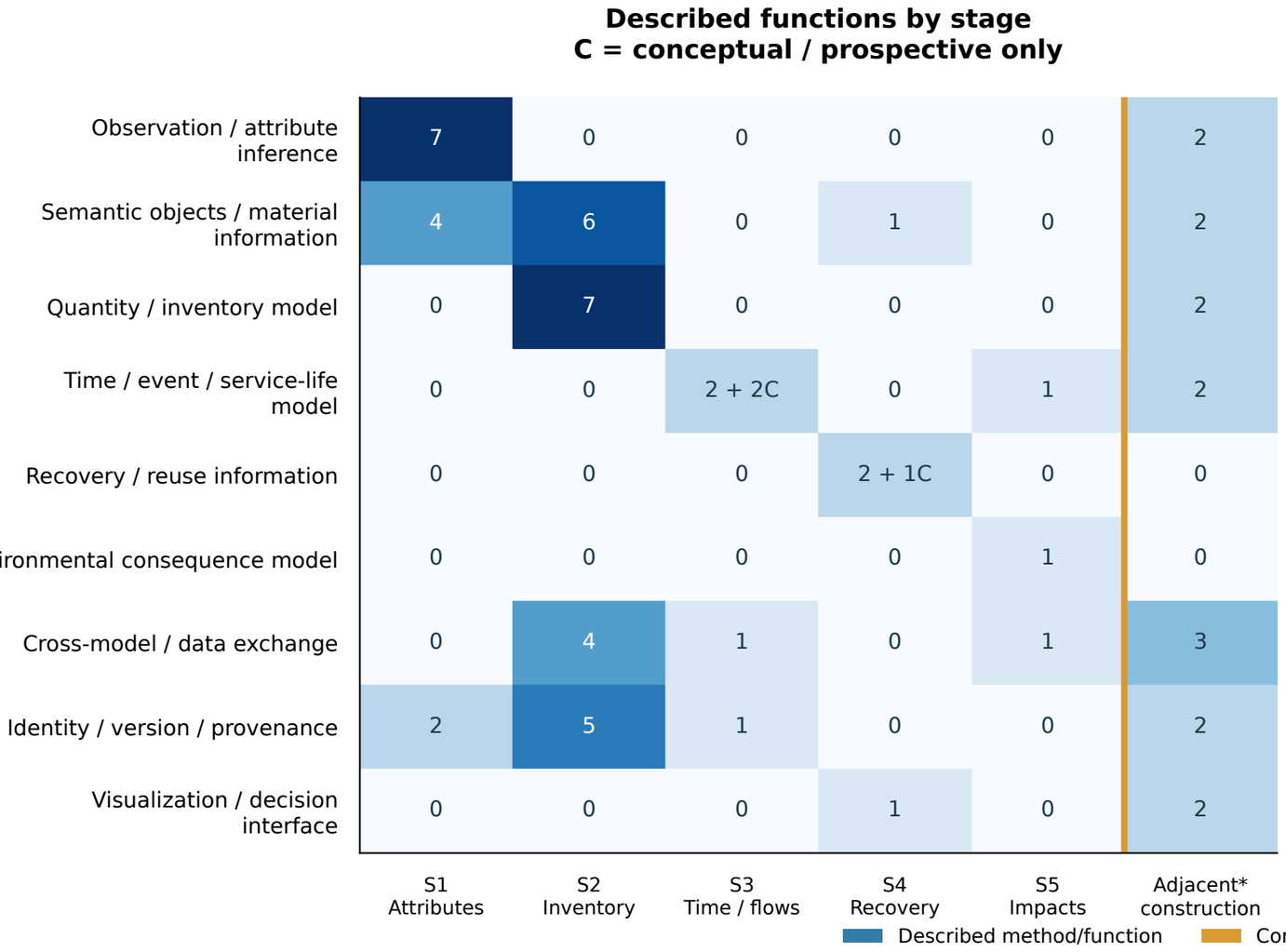
* Other → S5 keeps factor-, quantity- and damage-to-impact studies separate from recovery → impact.

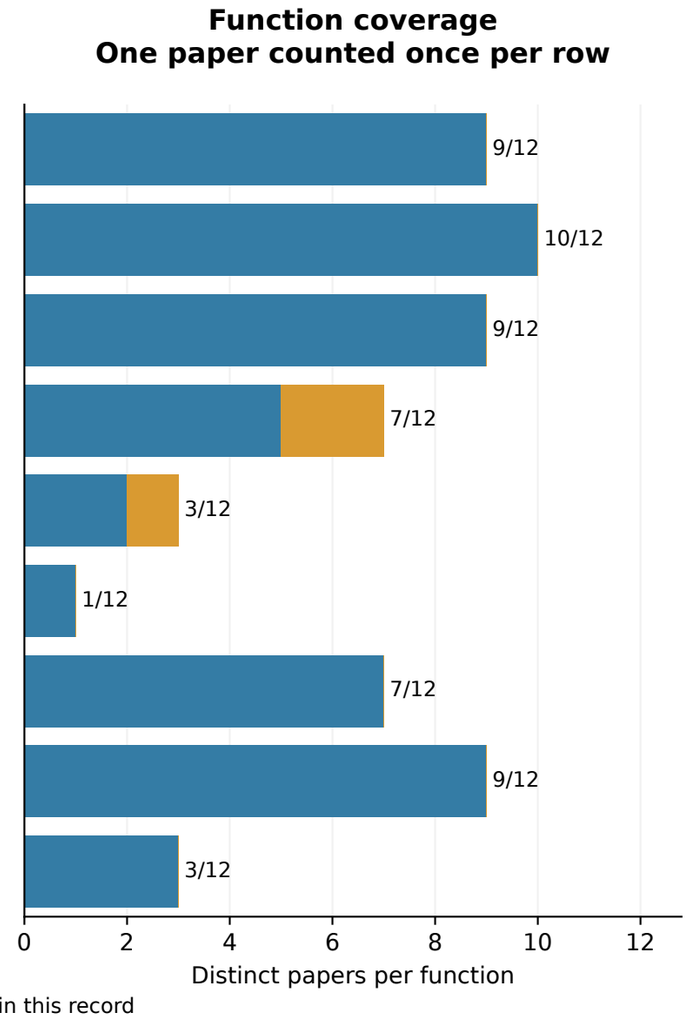
Twenty papers yield 23 context assessments. Zero means not recorded in this extraction; grey means no assessed context.

Quantified outputs for selected inputs do not establish a complete S1–S5 uncertainty chain.

02 Stages × Digital Twin and enabling functions

Twelve assessed domain / enabling studies. Functions are counted; this is not a count of twelve validated DTs.





Stage records with described functions: S1=7, S2=7, S3=2, S4=2, S5=1, X=3. Stage memberships overlap.

* Adjacent construction logistics, cost and schedule studies are not relabelled as building-stock turnover.

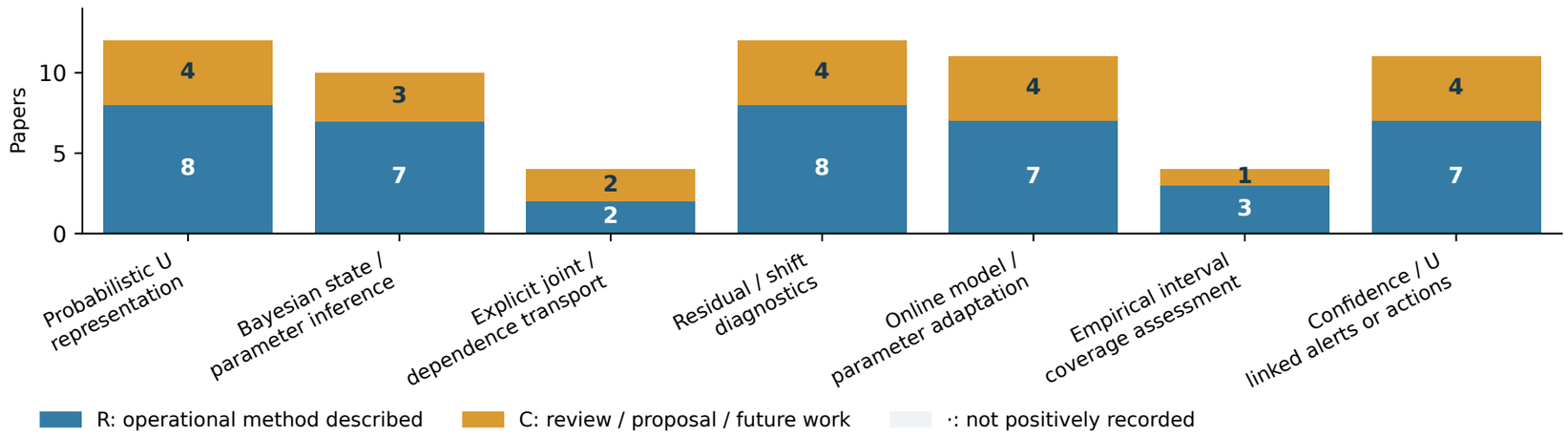
S3 includes retrospective event detection; S4 includes reuse information; neither label proves probabilistic flow/recovery modelling.

Zero = not positively coded, not proven absence. Sparse S4/S5 coverage is a property of this subset, not the entire literature.

03 Digital Twin architectures × uncertainty mechanisms

Thirteen archived appraisals. Method description, conceptual support and missing positive coding remain separate.

		Probabilistic U representation	Bayesian state / parameter inference	Explicit joint / dependence transport	Residual / shift diagnostics	Online model / parameter adaptation	Empirical interval coverage assessment	Confidence / U linked alerts or actions
Review / reference architecture	001 Physics-informed review	C	C	C	C	C	C	C
	009 Soft-sensor review	C	·	·	C	C	·	C
	010 Water-quality review	C	C	C	C	·	·	C
Hybrid model fusion	002 Hybrid fuel-system twin	R	R	·	R	R	·	R
Inference component	004 Conditional state–parameter inference	R	R	R	·	R	·	·
	008 Self-protection architecture	·	·	·	R	R	·	·
Feedback / control	015 Conflict-aware O-RAN control	R	·	·	R	R	·	R
	016 Parallel twins for NMPC	R	R	R	R	R	·	R
Layered monitoring	011 Bio360 event framework	C	C	·	R	·	·	R
	012 Field water-quality twin	R	R	·	R	C	R	R
	018 CNC tool-health twin	R	R	·	R	R	R	R
Bayesian learner	013 Hierarchical posterior evolution	R	R	·	R	R	R	R
	014 Adaptive physical-prior BNN	R	R	·	C	C	·	C

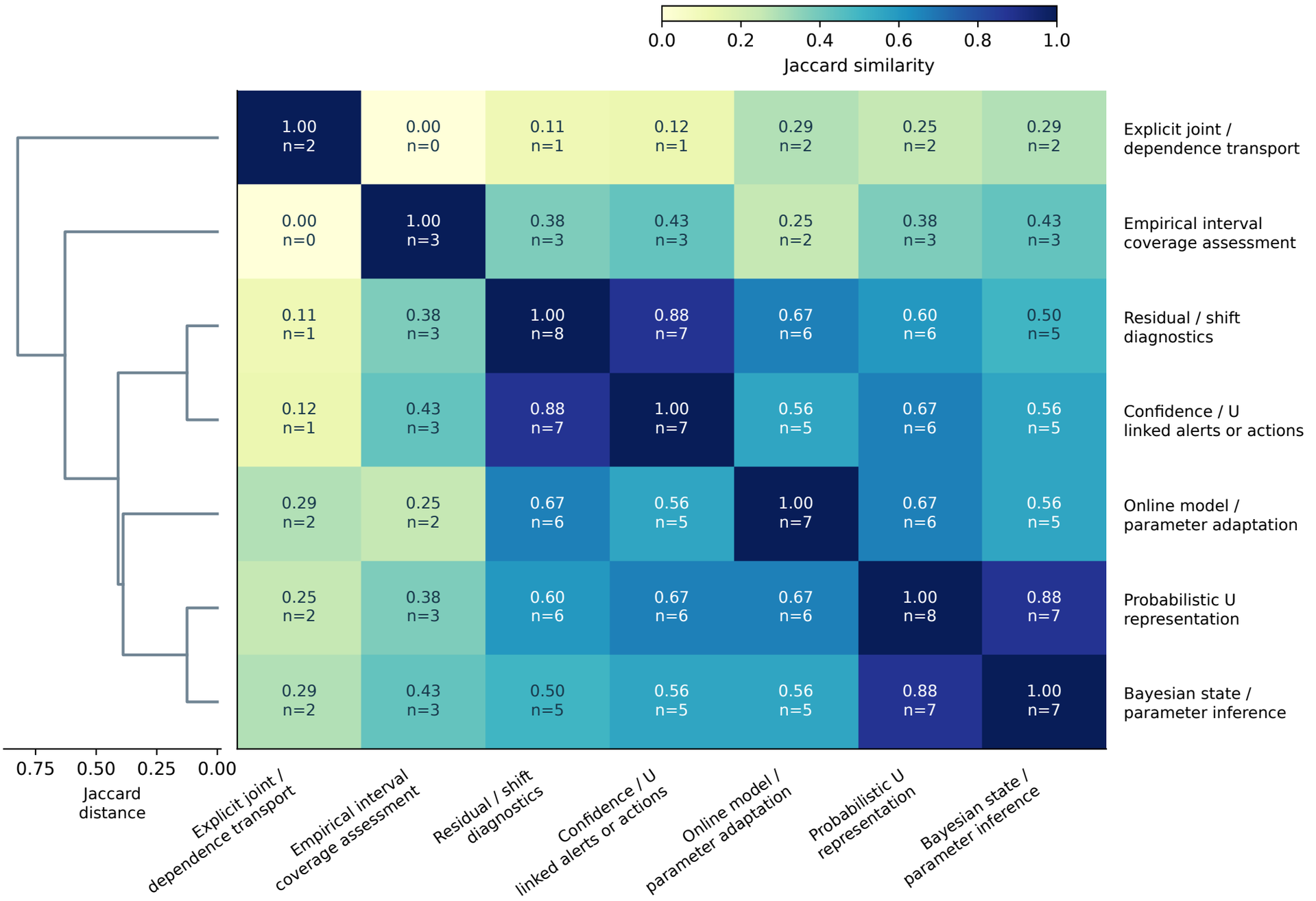


R does not mean independently validated. 004 is a preprint inference component; 015 is an adjacent control system.

Joint/dependence transport is coded for the specified variables only. It does not establish end-to-end DT continuity.

04 Which uncertainty mechanisms occur together?

Exploratory average-linkage clustering of ten method-level records. Reviews and conceptual-only tags are excluded.



Each cell shows similarity and the number of records reporting both methods. Clusters describe this selected working set.

Co-occurrence does not establish model coupling, causation or independent corroboration. Related author/study families remain a limitation.

Leave-one-record-out ranges are provided in the CSV. No significance test or stable architectural taxonomy is claimed.